Cholinergic Antagonists

PHA 201
Second-year Med

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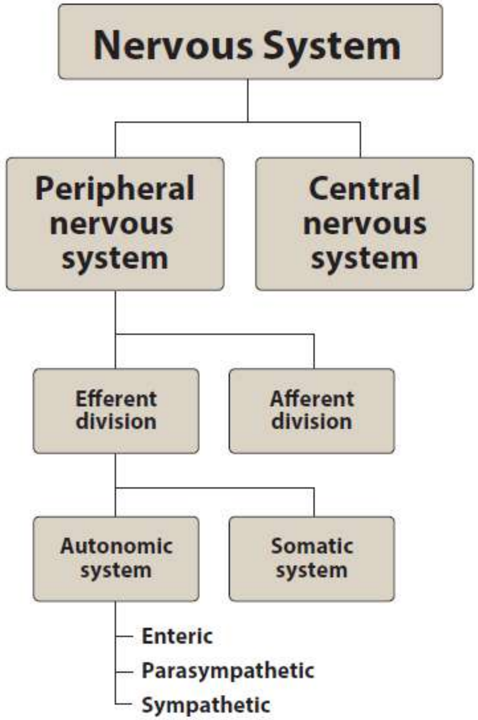


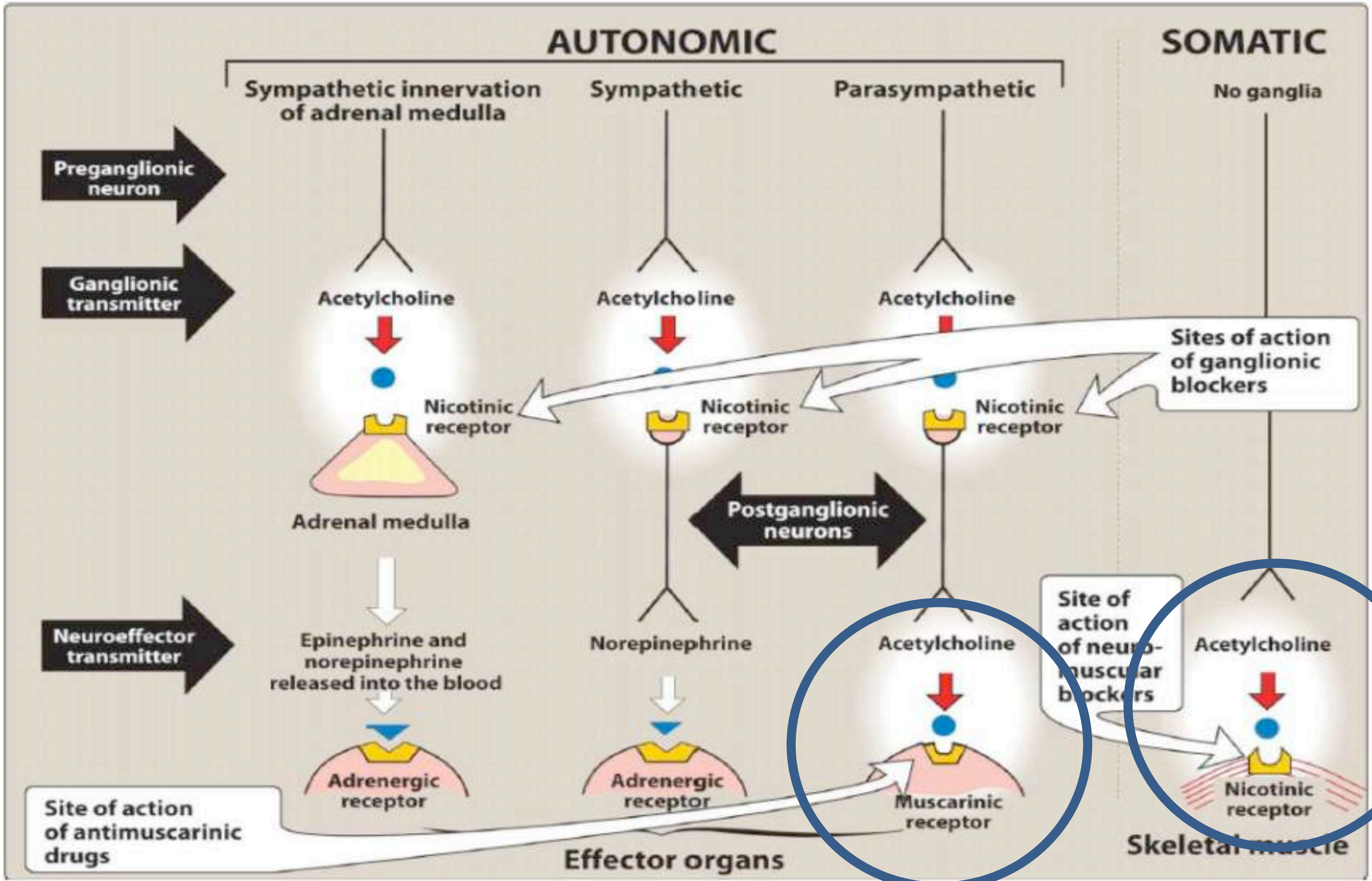
Objectives

At the end of the lecture, you should be able to:

1. Classify different cholinergic antagonists and their mechanism of actions.
2. Explain the pharmacological effects of atropine as a prototype antagonists on major organs.
3. Apply the pharmacological effects of cholinergic antagonists for its major clinical indications and contraindications.

Antimuscarinic agents

Atropine & Atropine Substitutes



Atropine

- It is well absorbed from GIT if given orally or from conjunctiva after ocular instillation & can cross BBB.
- Natural source: from ***Atropa belladona***

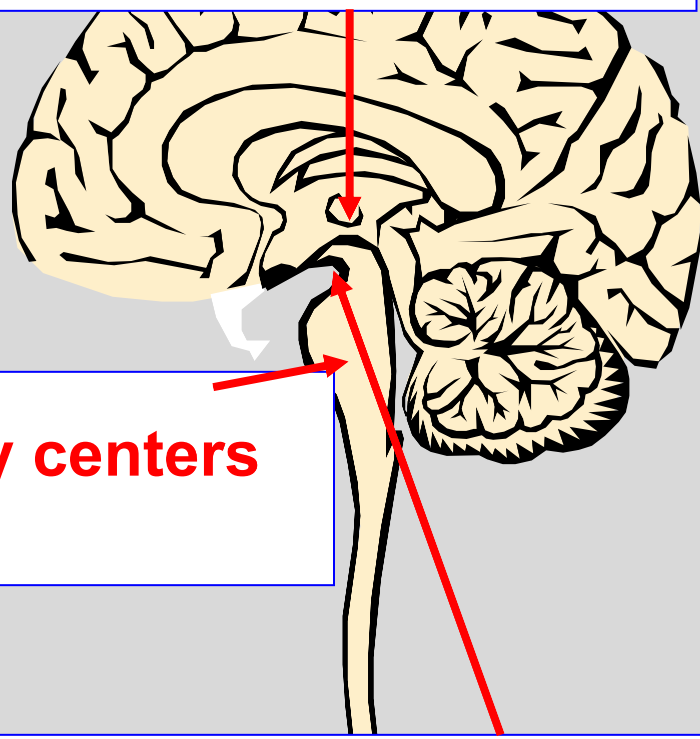
- **Mechanism of Action**

  Atropine causes **reversible** **competitive** blockade of the actions of ACh at muscarinic receptors (**non-selective for all muscarinic receptors**).

1. Atropine actions on CNS

❷ M_1 of basal ganglia

Anti-Parkinsonian action



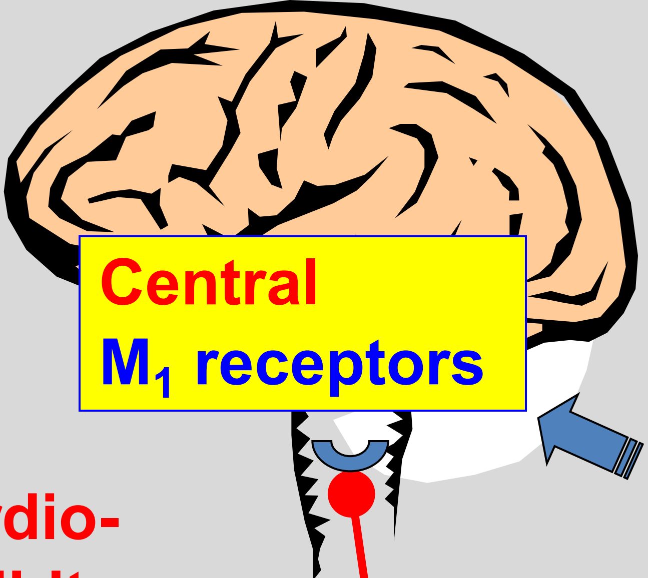
❸ M_2 of Medullary centers

RC stimulation

❶ M_1 of Chemoreceptive trigger zone

Antiemetic action

2. Atropine cardiac actions



Central
M_1 receptors

Cardio-
inhibitory
center

❶

Initial blockade of neuronal (presynaptic) M_1 receptors increases vagal Ach release
↓
Transient cardiac inhibition

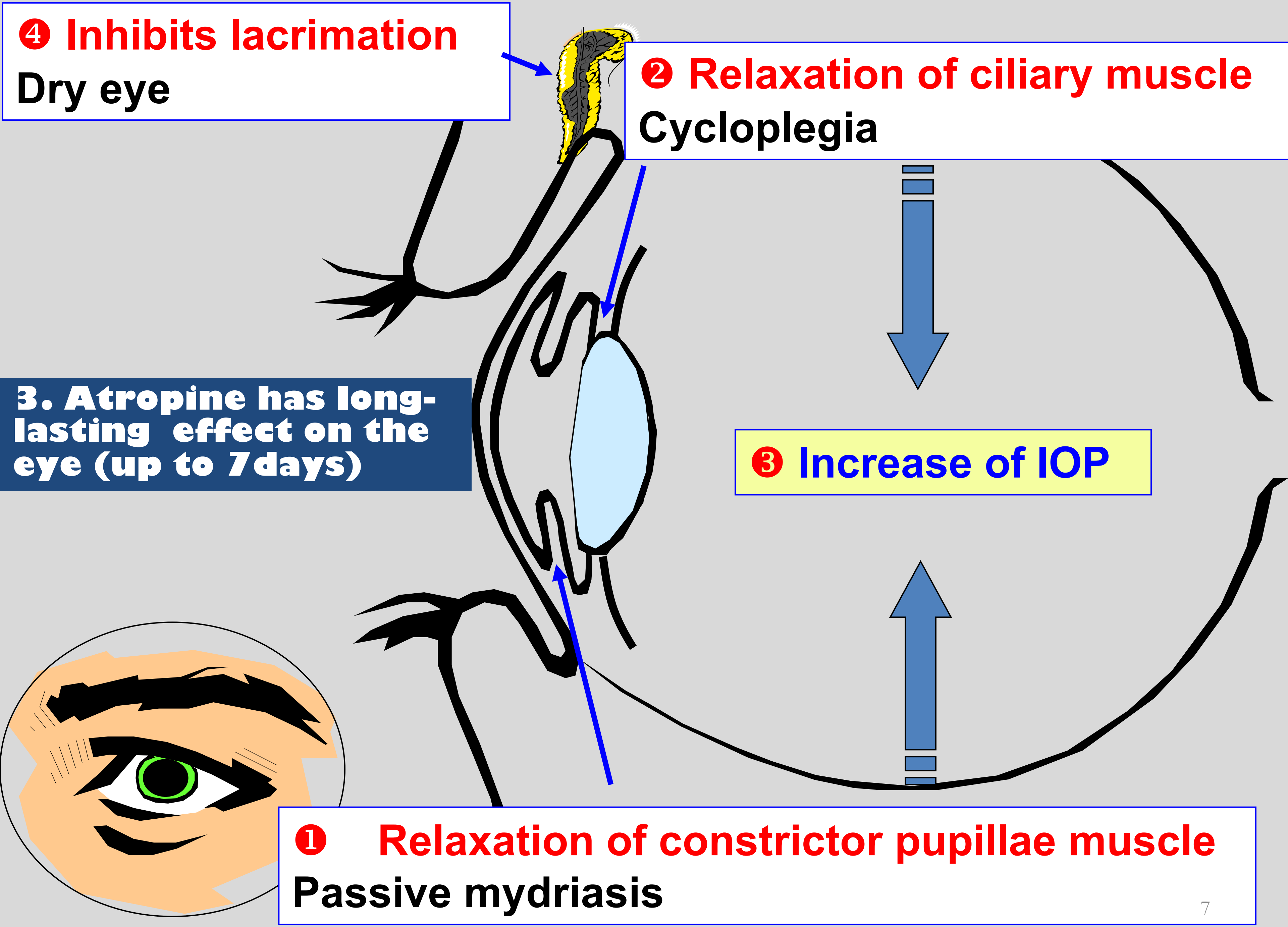
Presynaptic
M_1 receptors

Vagus nerve

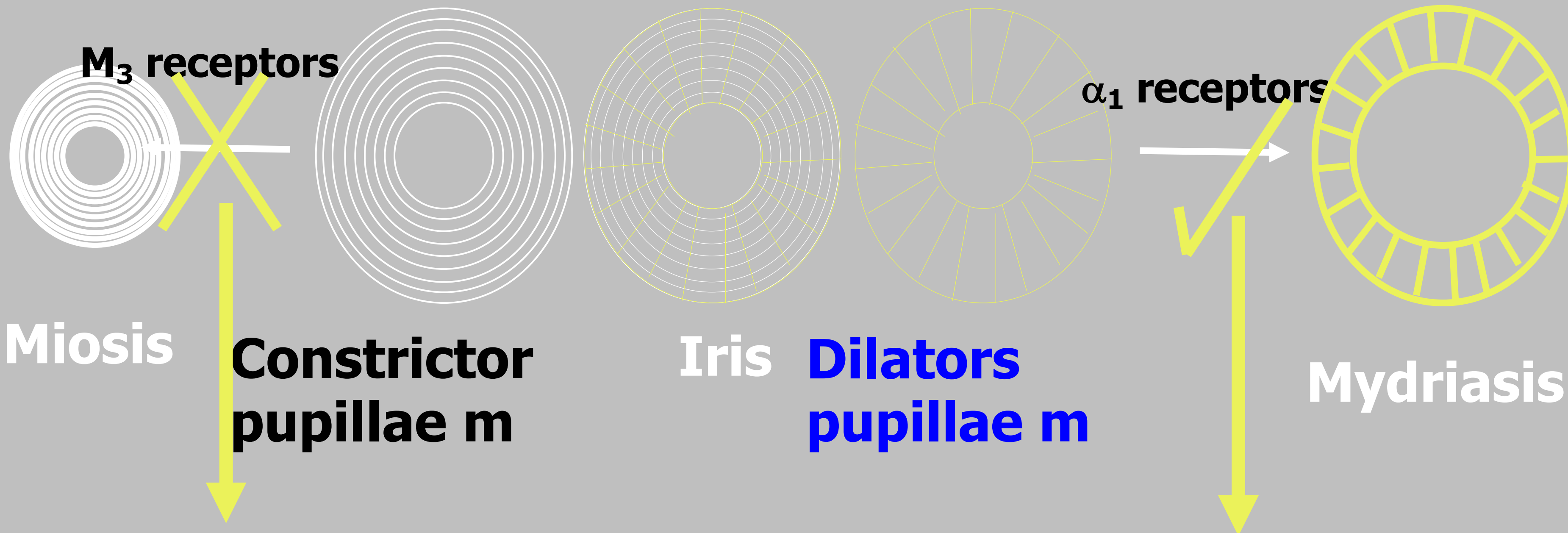
❷

Later blockade of cardiac M_2 receptors
↓
Dominant cardiac stimulation

Cardiac
M_2 receptors


❹ Inhibits lacrimation
Dry eye
❷ Relaxation of ciliary muscle
Cycloplegia
3. Atropine has long-lasting effect on the eye (up to 7days)
❸ Increase of IOP
❶ Relaxation of constrictor pupillae muscle
Passive mydriasis

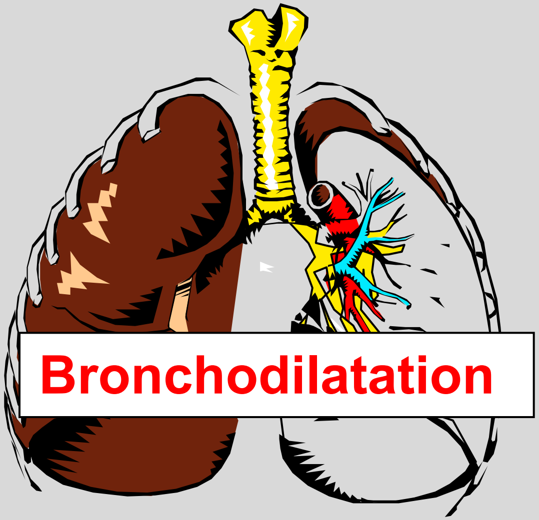
Passive vs Active mydriasis

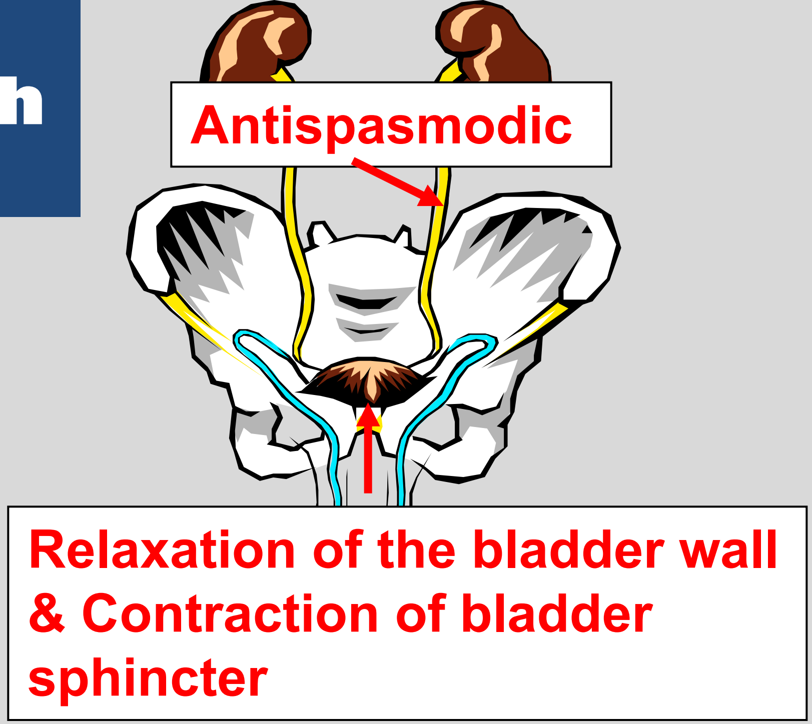


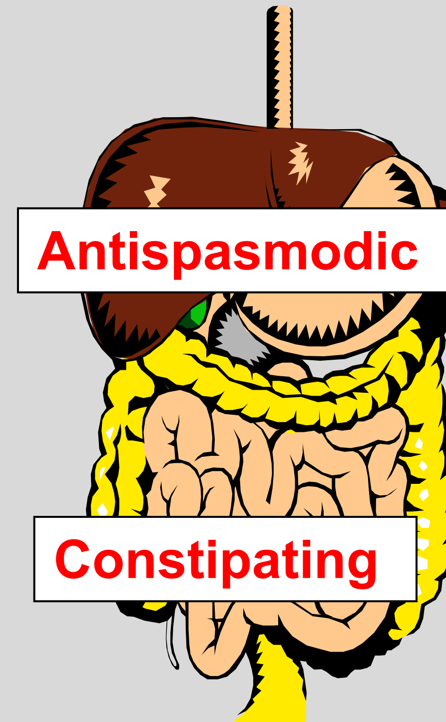
Anticholinergic drugs	**Sympathomimetic drugs**
Mydriasis by Relaxation of constrictor pupilae muscle	**Mydriasis by Contraction of dilator pupilae muscle**

Passive Mydriasis **Active Mydriasis**

4. Atropine & Other smooth muscles





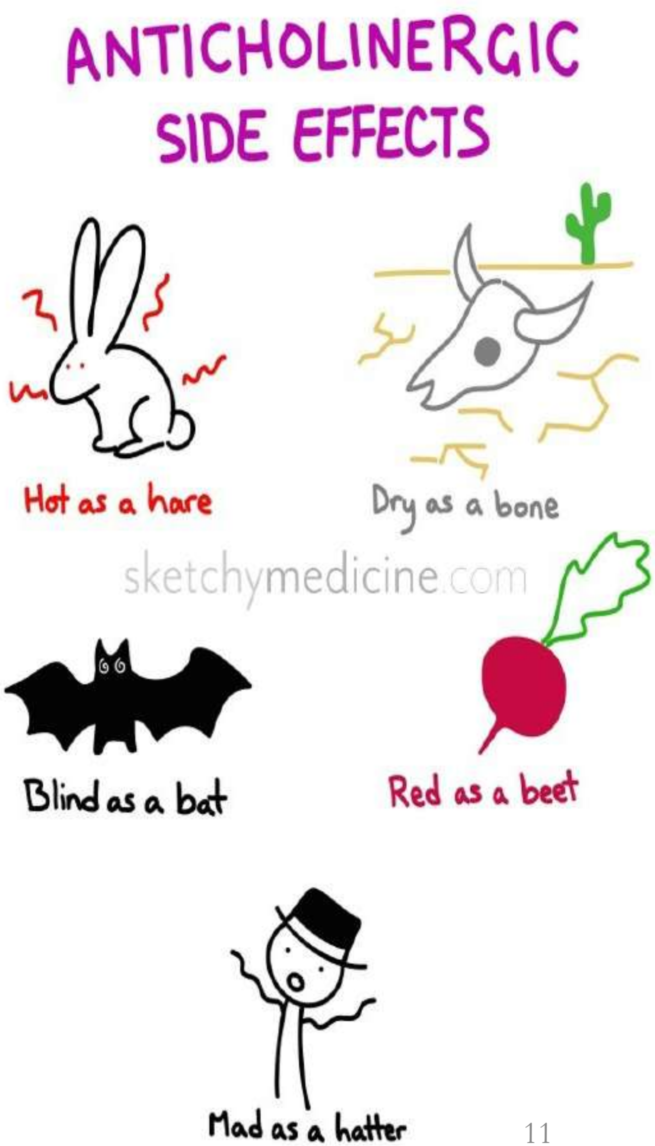


Atropine: Therapeutic Uses

1. **Preanesthetic medication** →inhibits secretions - dilates bronchi - anti-emetic - inhibits reflex bradycardia - stimulates respiration.
2. **Bradycardia**
3. **Organophosphate poisoning**
4. **Cycloplegic in children** (atropine is preferred to atropine substitutes in children as their cilliary muscle is strong & atropine substitutes are weaker cycloplegics than atropine).
5. **Travelers' diarrhea** (plus diphenoxylate, an opioid antidiarrheal) to enhance its constipating effect & to decrease its abuse.

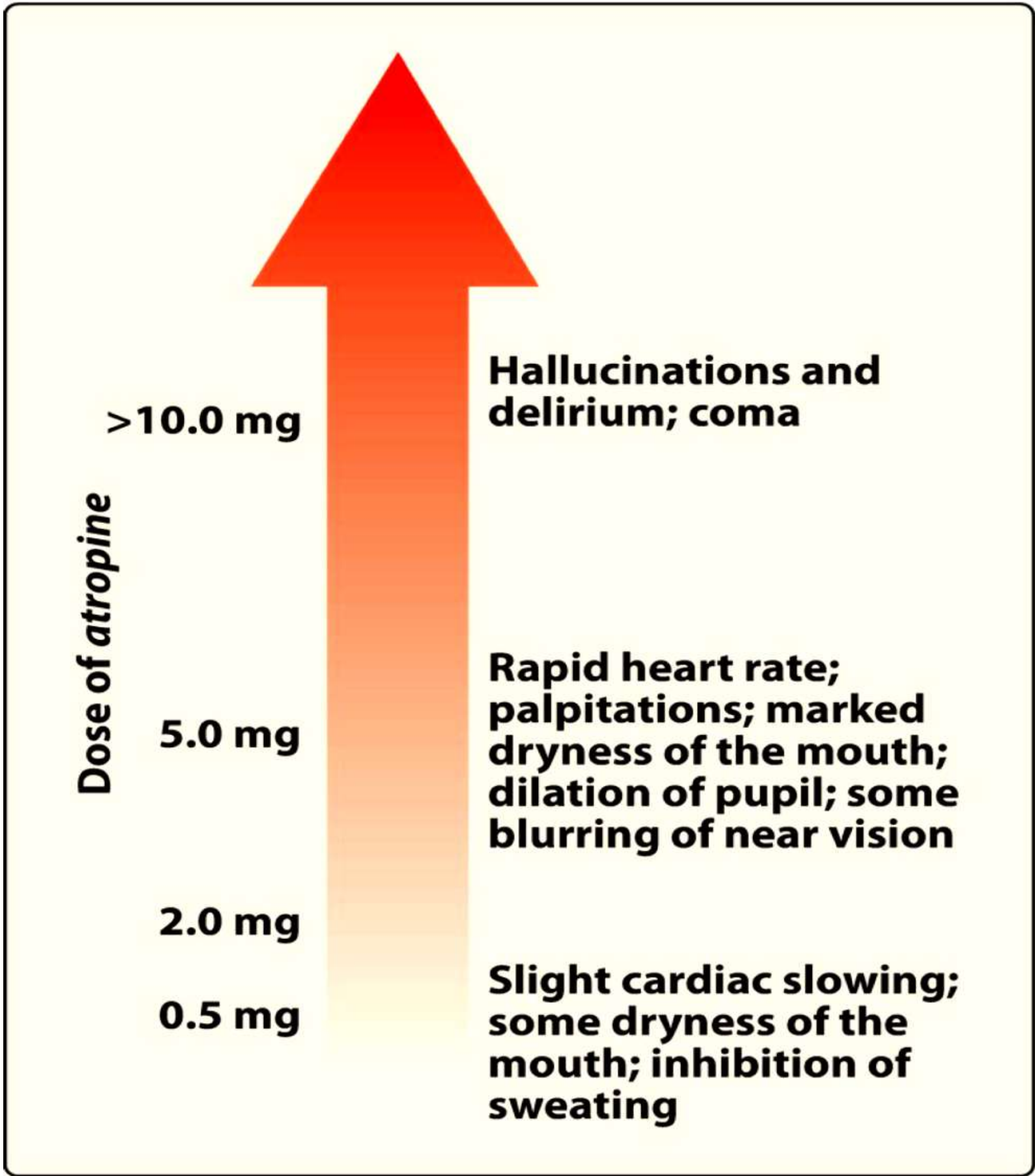
Atropine: Adverse Effects

1. Confusion, restlessness → hallucinations, delirium & mania→ ***Mad as a hatter.***
2. Dry eye - Blurred vision - photophobia→ ***Blind as a bat.***
3. Acute glaucoma in patients with narrow anterior chamber **(CI: glaucoma)**.
4. Dry mouth and skin →***Dry as a bone.***
5. Hyperthermia (complete skin dryness) →***Hot as a hare.***
6. Vasodilation & flushing →***Red as a beet.***
5. Tachycardia.
8. Urine retention in old patients with enlarged prostate **(CI: enlarged prostate)**.
9. Constipation.



Dose-dependent

Effects of Atropine



Atropine: Treatment of Poisoning

- Gastric lavage.
- Physostigmine (antagonizes central & peripheral effects).
- Diazepam: to control CNS excitement.
- Cooling blankets: to treat hyperthermia in children (not aspirin).

A 3-year-old child accidentally ingested a large quantity of atropine eye drops. In the emergency department, he presents with hyperthermia, dry mouth, dilated pupils, tachycardia, and hallucinations. Which of the following is the most appropriate treatment?

A) Pralidoxime

B) Physostigmine

C) Atropine

D) Neostigmine

A 48-year-old man presents to the ER after taking an unknown drug. He has dilated pupils, dry mucous membranes, tachycardia, urinary retention, and absent bowel sounds. His mental status is altered with hallucinations and agitation. What is the most likely diagnosis?

A) Organophosphate poisoning

B) Opioid overdose

C) Atropine toxicity

D) Beta-blocker overdose

ATROPINE SUBSTITUTES

Natural

- Scopolamine

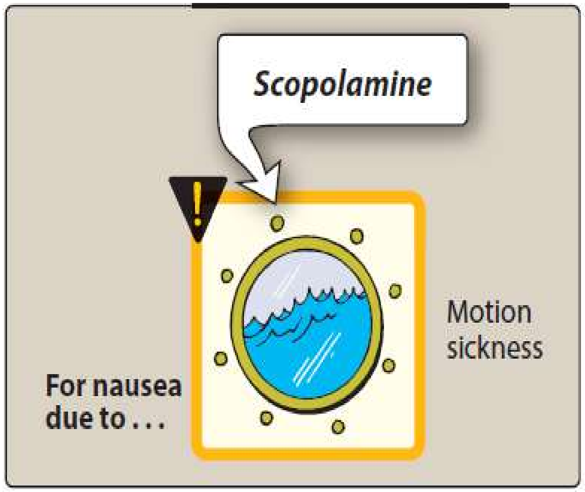
Synthetic

- Mydriatics
- Antispasmodics antisecretry
- Urinary substitutes
- Antiparkinson
- Bronchial substitutes

Natural Atropine Substitute
Scopolamine (Hyoscine) Uses

Scopolamine produces peripheral effects like those of atropine, but has greater action on the CNS than atropine

- **Mydriatic** (briefer than atropine).
- **Vomiting of motion sickness** (preferred as it is more effective than atropine).
- May induce CNS excitement & hallucination when used as preanesthetic in females → vivid dreams
- It may produce euphoria and is susceptible to **abuse**



A woman is prescribed an anticholinergic drug for motion sickness. She later experiences vivid dreams and hallucinations. Which of the following drugs was most likely used?

A) Atropine

B) Glycopyrrolate

C) Scopolamine

D) Dicyclomine

Synthetic Atropine Substitute
1. Mydriatic cycloplegics

Cyclopentolate - Tropicamide – Homatropine

used for:

- Iridocyclitis; alternating with miotics to prevent synechia (**homatropine**).
- Measuring refractive errors & for fundus examination.

N.B. : *They are shorter acting than atropine,* ***Tropicamide*** *produces mydriasis for* ***6 hours*** *and* ***Cyclopentolate*** *for* ***24 hours*** → *action is easier to reverse → preferred to atropine (except in children).*

A child with acute iridocyclitis is prescribed a mydriatic-cycloplegic agent. Which of the following drugs is preferred due to its shorter duration of action?

A) Atropine

B) Tropicamide

C) Scopolamine

D) Glycopyrrolate

Synthetic Atropine Substitute
2. Antispasmodics Antisecretory

4ry drugs→ less absorption → less CVS & CNS side effects)

- **Glycopyrrolate**: anti-secretory in preanaesthetic medication.
- **Hyoscine butylbromide**: antispasmodic in renal, biliary & intestinal colic & in irritable bowel syndrome.

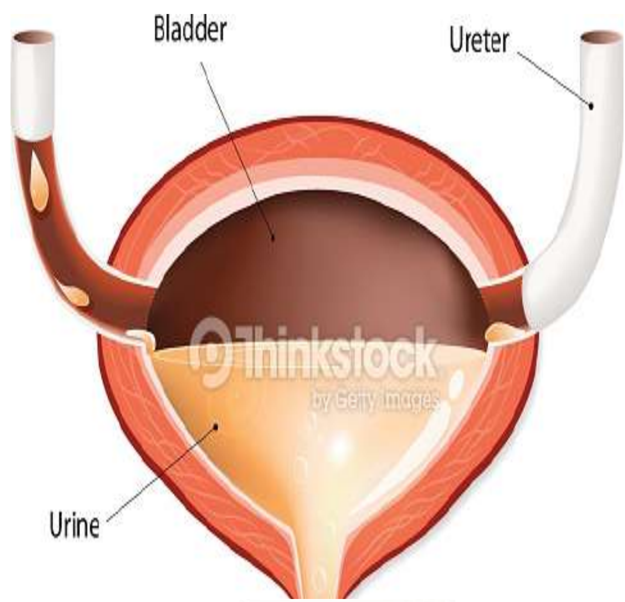
3ry drugs: Dicyclomine & Pirenzepine:

Selective M1 blockers → decrease gastric HCL

Synthetic Atropine Substitute
3. Urinary Substitutes

Solifenacin, Darifenacin and Oxybutynin:

- They induce bladder wall relaxation & sphincter contraction.
- More uroselective than atropine with fewer adverse effects (no dry mouth & blurred vision)
- Used in nocturnal enuresis & in urge incontinence.



A 60-year-old male patient with urinary urgency and incontinence is prescribed a selective M3 antagonist. Which of the following drugs is most appropriate for this condition?

A) Ipratropium

B) Scopolamine

C) Solifenacin

D) Pancuronium

Synthetic Atropine Substitute
4. Bronchial Substitutes

A. Ipratropium (non selective M2 / M3 blocker)

Inhaled bronchodilator (M3 blocker) → no systemic atropine side effects.

- Used in asthma & COPD: delayed onset compared to beta 2 agonists.
- Tolerance develops due to block of vagal presynaptic M2 receptor →↑ ACh.

B. Tiotropium (selective M3 blocker)

- Longer acting than ipratropium → used once/d; for maintenace in COPD.
- Does not block M2 receptors → no tolerance.

A patient with chronic obstructive pulmonary disease (COPD) is prescribed an inhaled anticholinergic agent. After prolonged use, <u>he does not develop tolerance due to the drug's selective mechanism. Which drug is most likely prescribed?</u>

A) Tiotropium

B) Ipratropium

C) Succinylcholine

D) Benztropine

Synthetic Atropine Substitute
5. Antiparkinsonian

Benztropine, Biperiden, Benzhexol:

Used in drug-induced Parkinsonism: preferred to dopaminergic drugs.

Adjuvants in Parkinsonism presenting with tremors & to control sialorrhea.

A 70-year-old male with Parkinson's disease is experiencing excessive salivation and tremors. His physician prescribes an anticholinergic medication that crosses the blood-brain barrier and selectively acts on the central nervous system. <u>Which of the following drugs is most appropriate for this patient?</u>

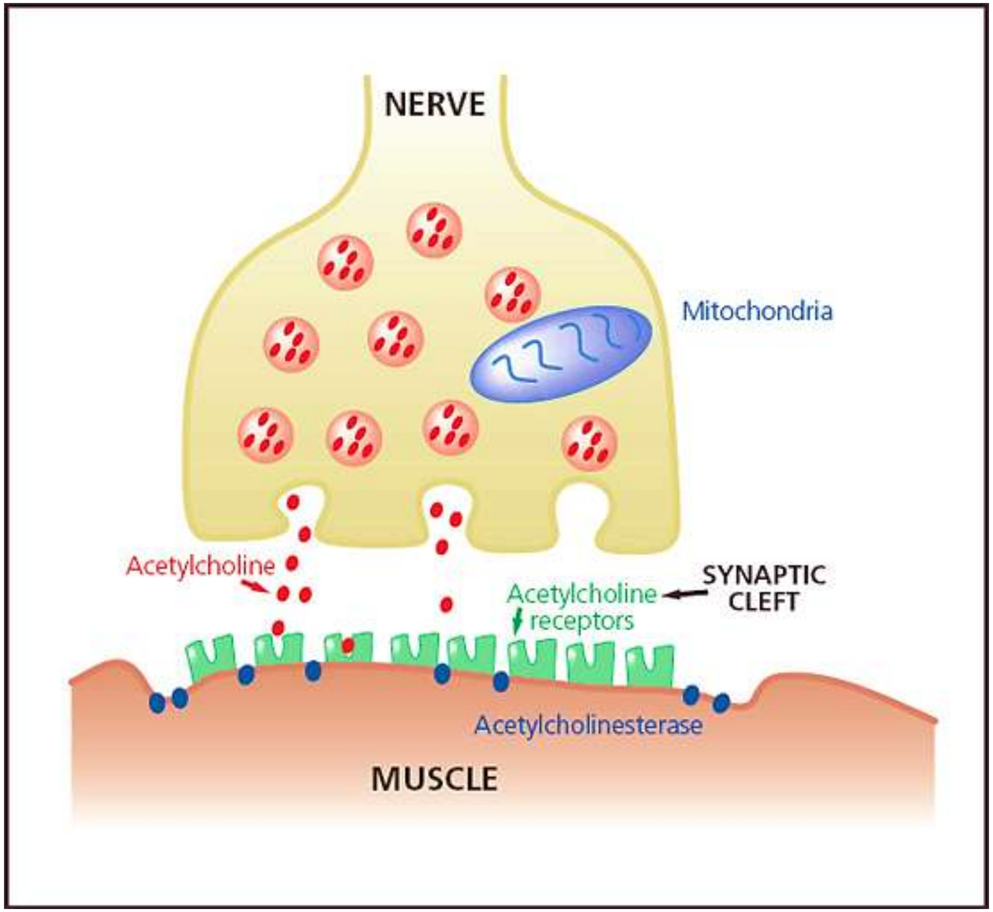
A) Glycopyrrolate

B) Benztropine

C) Ipratropium

D) Oxybutynin

NEUROMUSCULAR BLOCKERS



Neuromuscular Blockers

- Classify neuromuscular blocking agents based on their mechanism of action (non-depolarizing vs. depolarizing)
- Explain the unique properties of succinylcholine, including its rapid onset, mechanism of action, and clinical applications
- Identify important adverse effects of NMBs, including malignant hyperthermia, hyperkalemia, and residual neuromuscular blockade
- Describe drug interactions that potentiate or antagonize

Neuromuscular Blockers

Classification of neuromuscular blocking agents (two major classes):

1. NON-DEPOLARIZING (Competitive) AGENTS

Mechanism: • Competitive antagonists at nicotinic receptors • Block ACh binding without activating receptor • Prevent depolarization and muscle contraction

Key Characteristics: • No initial fasciculations • Reversible with anticholinesterases (↑ ACh concentration) • Dose-dependent duration

Clinical Examples: • Short-acting: Mivacurium (15-20 min) • Intermediate-acting: Rocuronium, vecuronium, atracurium, cisatracurium (30-40 min) • Long-acting: Pancuronium (60-90 min)

2. DEPOLARIZING AGENT

Mechanism: • Agonist at nicotinic receptors • Causes sustained depolarization • Receptor desensitization → blockade

Key Characteristics: • Initial muscle fasciculations (uncoordinated twitching) • NOT reversible with anticholinesterases • Very short duration (5-10 min)

Clinical Example: • Succinylcholine (ONLY depolarizing agent in clinical use)

Feature	Non-Depolarizing	Depolarizing
Mechanism	Competitive antagonism	Agonist → desensitization
Initial effect	No fasciculations	Fasciculations
Onset	Variable (1-5 min)	Very rapid (30-60 sec)
Duration	Variable (20-90 min)	Very short (5-10 min)
Reversal	Anticholinesterases ✓	Anticholinesterases ✗
Examples	Rocuronium, vecuronium, others	Succinylcholine only

CLINICAL APPLICATIONS:

✓ Facilitate endotracheal intubation

✓ Provide muscle relaxation during surgery

✓ Optimize surgical field visualization and access

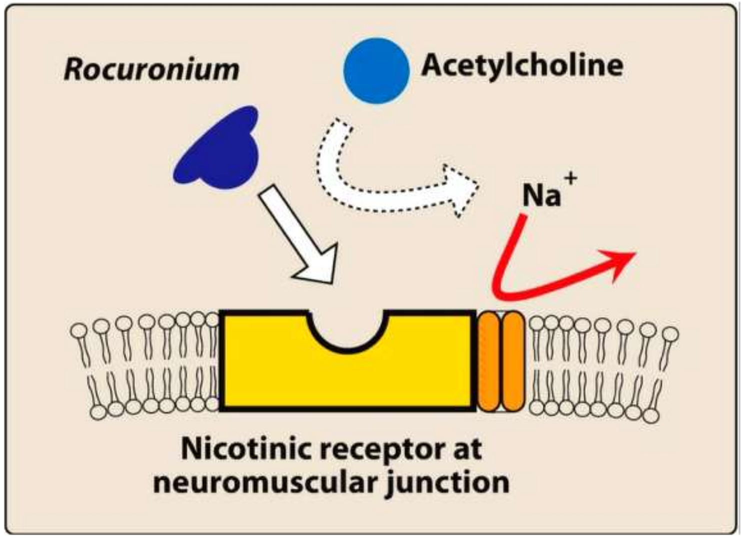
✓ Enable mechanical ventilation in ICU

✓ Reduce anesthetic agent requirements

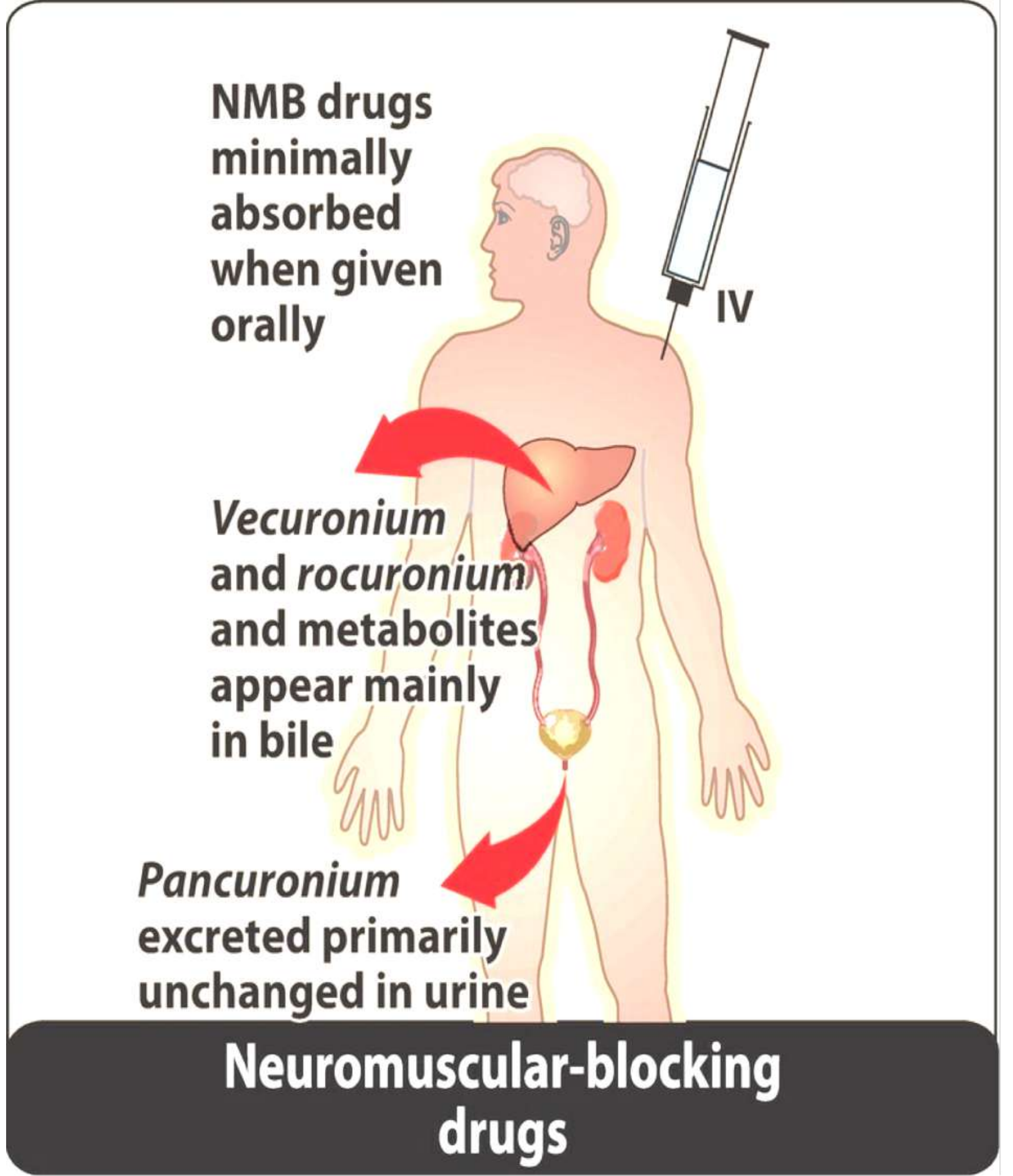
✓ Prevent patient movement during delicate procedures

Competitive NMBs (CNMB)

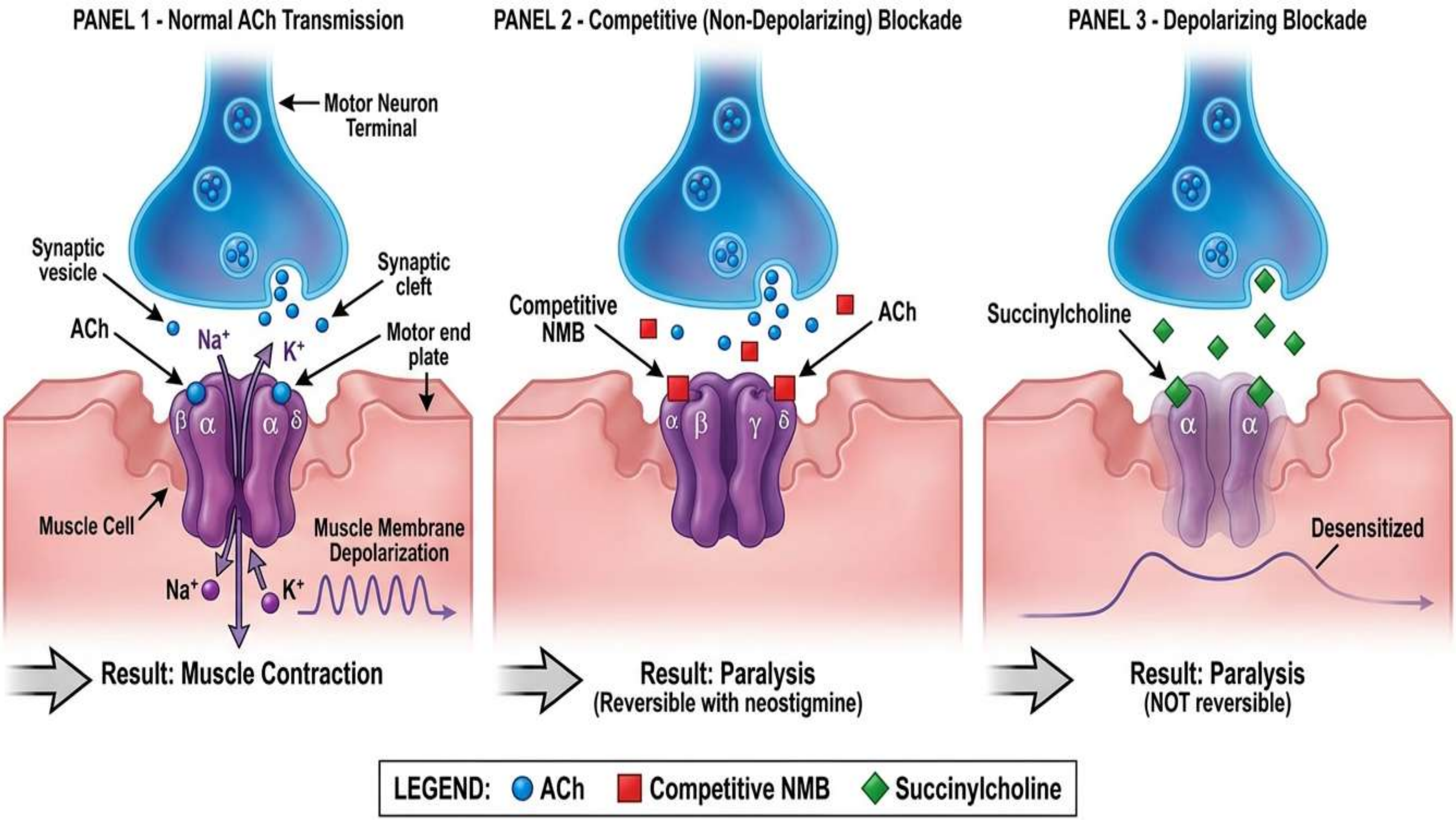
- "**Curare**" is the first known member. It is a mixture of naturally occurring alkaloids having neuromuscular blocking activity of which "**Tubocurarine**" is the most important one.
- It has been replaced by agents with fewer adverse effects, such as: pancuronium, rocuronium, and vecuronium
- They are injected IV because their oral absorption is minimal (4ry amines)
- They do not cross BBB



Pharmacokinetics



Neuromuscular Junction and Receptor Mechanisms



Competitive NMBs (CNMB)

- **Mechanism of action:**
  - **At low doses:**
    - They competitively block the Nm → muscle relaxation
    - Their action can be overcome by increasing the concentration of Ach in the synaptic gap e.g. anticholinesterase such as neostigmine & pyridostigmine
    - Anesthesiologist often employ this strategy to shorten the duration of neuromuscular blockade
  - **At high doses:**
    - Block the ion channel at the end plate i.e. can not be reversed by anticholinesterase

CNMB Effects

- **They produce paralysis of body muscles in the following sequence:**
  1. Eye Muscles (double vision)
  2. Face Small muscles
  3. Pharynx (difficulty in swallowing)
  4. Limbs
  5. Respiratory Muscles (the last to be affected)

CNMB Drug Interactions

1. Cholinesterase inhibitors
   e.g. neostigmine: antagonism

2. Halogenated hydrocarbon anesthetics
   e.g. isoflurane, sevoflurane: synergism

3. Aminoglycosides antibiotics
   e.g. gentamycin or tobramycin: synergism
4. Calcium channel blockers: (e.g., verapamil, diltiazem)
   - Mechanism: ↓ Calcium-dependent ACh release
   - Clinical significance: Dose adjustment may be needed

REVERSAL STRATEGIES for NON-DEPOLARIZING (competitive) AGENTS

NEOSTIGMINE (Traditional Reversal Agent)

Mechanism: • Acetylcholinesterase inhibitor

• ↑ ACh in synaptic cleft

• Increased ACh competes with non-depolarizing blocker

• Restores neuromuscular transmission

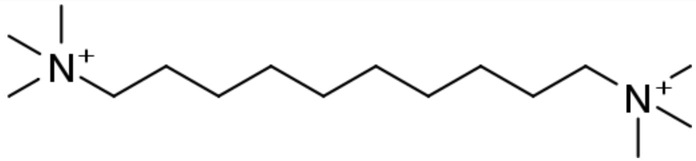
Dosing: • 0.04-0.07 mg/kg (max 5 mg) • MUST give with antimuscarinic agent:

Glycopyrrolate 0.01 mg/kg (preferred - longer duration)

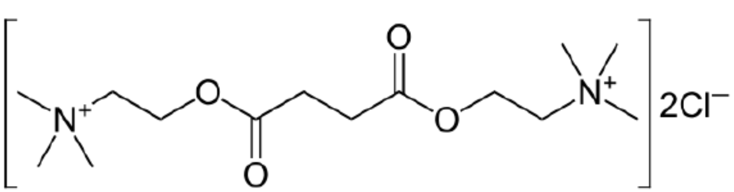
Atropine 0.02 mg/kg (faster onset)

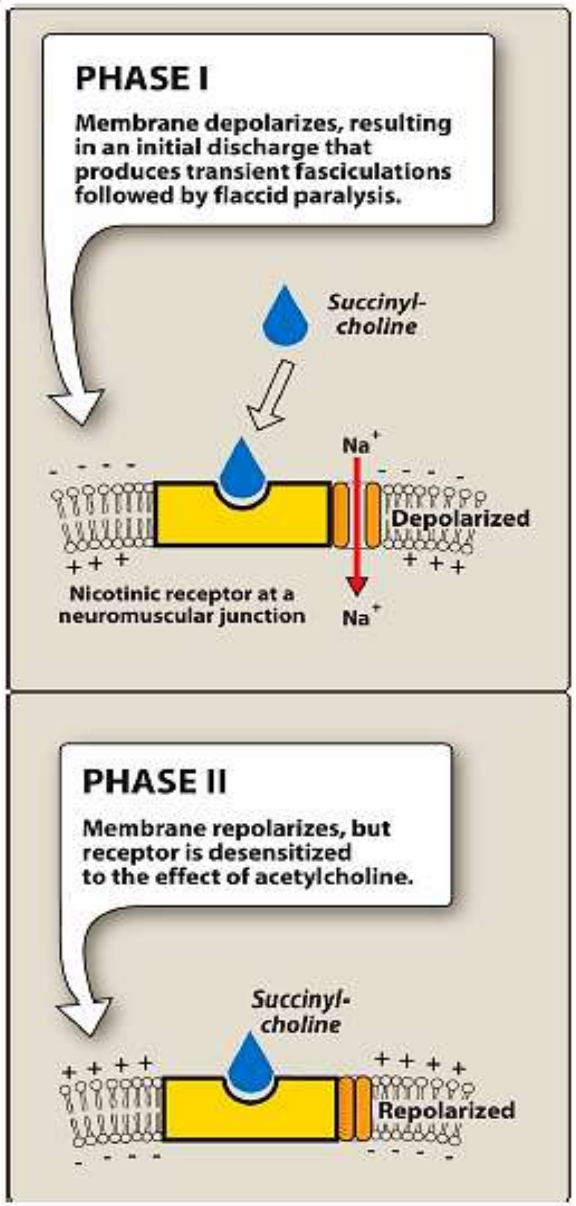
Why antimuscarinic needed? • Neostigmine inhibits ALL cholinesterase (not selective) • ↑ ACh at muscarinic receptors → bradycardia, bronchospasm, salivation • Antimuscarinic blocks these unwanted effects

Depolarizing NMBs (DNMB)



- They are biquaternary ammonium compounds e.g. decamethonium & succinylcholine.
- Suxamethonium (succinylcholine) consists of two Ach molecules linked by their acetyl groups, so its action is shorter than decamethonium because it is quickly hydrolysed by plasma pseudocholinesterase.





DNMBs actions & uses

- They produce initial short-lasting muscle fasciculation, followed within few minutes by paralysis.
- "Rapid, non-selective paralysis of all muscle groups within 30-60 seconds"
- The drug does not produce a ganglionic block except in high doses.
- They have weak histamine releasing action (allergy and bronchospasm).

Uses

- Because of its rapid onset and short duration of action, succinylcholine is useful when rapid endotracheal intubation is required.
- It is also used during electroconvulsive shock treatment.

DNMBs adverse effects

- Prolonged paralysis and apnea due to genetic deficiency of Pseudocholinesterase
- Postoperative myalgia, particularly, neck, back and abdomen.
- Malignant hyperthermia (an intense muscle spasm and a very sudden rise in body temperature). "Succinylcholine ALONE can trigger malignant hyperthermia in genetically susceptible patients. Volatile anesthetics are separate triggers."
- Hyperkalemia.

HYPERKALEMIA ⚠️ MOST DANGEROUS ADVERSE EFFECT

MECHANISM:

- Upregulation of extrajunctional ACh receptors
- Massive K+ release with depolarization
- ↑ K+ by 5-10 mEq/L (can cause cardiac arrest)

HIGH-RISK CONDITIONS (CONTRAINDICATIONS):

- Burns >24 hours old
- Spinal cord injury >24 hours
- Prolonged immobilization
- Denervation injuries
- Neuromuscular diseases
- Severe infections

TIMING: Risk period = 24 hours to 1 year post-injury

CONSEQUENCE: Sudden cardiac arrest possible

A 45-year-old male undergoes surgery and is given an IV neuromuscular blocker for muscle relaxation. The drug causes an initial phase of muscle fasciculations followed by paralysis. <u>Which of the following drugs was most likely administered?</u>

A) Pancuronium

B) Succinylcholine

C) Rocuronium

D) Vecuronium

A patient is undergoing surgery and is administered a neuromuscular blocker. After surgery, the anesthesiologist administers neostigmine to reverse the blockade. Which of the following drugs was most likely used as the initial neuromuscular blocker?

A) Succinylcholine

B) Pancuronium

C) Atropine

D) Tiotropium

A 55-year-old patient undergoing surgery is given a neuromuscular blocking agent that causes histamine release, ganglionic blockade, and prolonged apnea in patients with pseudocholinesterase deficiency. Which drug is most likely responsible?

A) Rocuronium

B) Succinylcholine

C) Pancuronium

D) Vecuronium

A 40-year-old male is given a neuromuscular blocker for intubation. Within minutes, he develops malignant hyperthermia, characterized by severe muscle rigidity and hyperthermia. Which of the following drugs was most likely responsible?

A) Succinylcholine

B) Atracurium

C) Rocuronium

D) Vecuronium